

Keys Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE, ENROLLMENT.

AIM: To demonstrate PRIMARY KEY, UNIQUE KEY, FOREIGN KEY, and COMPOSITE KEY constraints and observe how they enforce data integrity.

-- 1. PRIMARY KEY: Uniquely identifies each row

```
-- Already applied earlier:
-- ALTER TABLE STUDENT ADD PRIMARY KEY (RegNo);

-- Try inserting a duplicate RegNo (should fail)
INSERT INTO STUDENT VALUES (191, 'Duplicate', 'M', '2001-01-01', 9999999999,
'Chennai');
```

```
mysql> -- Already applied earlier:
mysql> -- ALTER TABLE STUDENT ADD PRIMARY KEY (RegNo);
mysql>
mysql> -- Try inserting a duplicate RegNo (should fail)
mysql> INSERT INTO STUDENT VALUES (191, 'Duplicate', 'M', '2001-01-01', 9999999999, 'Chennai');
ERROR 1062 (23000): Duplicate entry '191' for key 'student.PRIMARY'
mysql>
mysql> |
```

-- 2. UNIQUE KEY: Prevents duplicate values, allows one NULL

```
-- Already applied earlier:
-- ALTER TABLE STUDENT ADD CONSTRAINT uq_student_name UNIQUE (Name);

-- Try inserting a duplicate Name (should fail)
INSERT INTO STUDENT VALUES (196, 'Aravind Kumar', 'M', '2001-05-05', 9884099999,
'Chennai');
```

```
mysql> -- Try inserting a duplicate Name (should fail)
mysql> INSERT INTO STUDENT VALUES (196, 'Aravind Kumar', 'M', '2001-05-05', 9884099999, 'Chennai');
ERROR 1062 (23000): Duplicate entry 'Aravind Kumar' for key 'student.uq_student_name'
mysql> |
```

-- 3. FOREIGN KEY: Enforces a link to a parent table's key

```
-- Already applied earlier:
-- ALTER TABLE COURSE ADD CONSTRAINT fk_course_faculty
-- FOREIGN KEY (FacNo) REFERENCES FACULTY(FacNo);

-- Try inserting a course with a FacNo that doesn't exist in FACULTY (should fail)
INSERT INTO COURSE VALUES ('C99', 'Ghost Course', 'T', '1', 'H999', 'F999');
```

```
mysql> -- Try inserting a course with a FacNo that doesn't exist in FACULTY (should fail)
mysql> INSERT INTO COURSE VALUES ('C99', 'Ghost Course', 'T', '1', 'H999', 'F999');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('lab`.`course`, CONSTRAINT `fk_course_faculty` FOREIGN KEY (`FacNo`) REFERENCES `faculty` (`FacNo`))
mysql>
```

-- 4. COMPOSITE KEY: Primary key made of more than one column

```
CREATE TABLE ENROLLMENT (
    RegNo      INT,
    CourseNo   VARCHAR(3),
    EnrollDate DATE,
    PRIMARY KEY (RegNo, CourseNo),
    FOREIGN KEY (RegNo) REFERENCES STUDENT(RegNo),
    FOREIGN KEY (CourseNo) REFERENCES COURSE(CourseNo)
);
```

```
mysql> CREATE TABLE ENROLLMENT (
->     RegNo      INT,
->     CourseNo   VARCHAR(3),
->     EnrollDate DATE,
->     PRIMARY KEY (RegNo, CourseNo),
->     FOREIGN KEY (RegNo) REFERENCES STUDENT(RegNo),
->     FOREIGN KEY (CourseNo) REFERENCES COURSE(CourseNo)
-> );
ERROR 1050 (42S01): Table 'enrollment' already exists
```

```
INSERT INTO ENROLLMENT VALUES (191, 'C01', '2026-06-01');
INSERT INTO ENROLLMENT VALUES (191, 'C02', '2026-06-01');    -- allowed: different CourseNo
INSERT INTO ENROLLMENT VALUES (191, 'C01', '2026-07-01');    -- fails: same (RegNo, CourseNo) pair
```

```
mysql> INSERT INTO ENROLLMENT VALUES (191, 'C01', '2026-06-01');
ERROR 1062 (23000): Duplicate entry '191-C01' for key 'enrollment.PRIMARY'
mysql> INSERT INTO ENROLLMENT VALUES (191, 'C02', '2026-06-01');    -- allowed: different CourseNo
ERROR 1062 (23000): Duplicate entry '191-C02' for key 'enrollment.PRIMARY'
mysql> INSERT INTO ENROLLMENT VALUES (191, 'C01', '2026-07-01');    -- fails: same (RegNo, CourseNo) pair
ERROR 1062 (23000): Duplicate entry '191-C01' for key 'enrollment.PRIMARY'
mysql> |
```

-- 5. Viewing key structure with DESC / SHOW KEYS

```
DESC ENROLLMENT;  
SHOW KEYS FROM ENROLLMENT;
```

```
mysql> DESC ENROLLMENT;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
CourseNo	varchar(3)	NO	PRI	NULL	
EnrollDate	date	YES		NULL	

3 rows in set (0.05 sec)

```
mysql> SHOW KEYS FROM ENROLLMENT;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null
Index_type	Comment	Index_comment	Visible	Expression					
enrollment	0	PRIMARY	1	RegNo	A	4	NULL	NULL	
BTREE			YES	NULL					
enrollment	0	PRIMARY	2	CourseNo	A	5	NULL	NULL	
BTREE			YES	NULL					
enrollment	1	CourseNo	1	CourseNo	A	4	NULL	NULL	
BTREE			YES	NULL					

3 rows in set (0.02 sec)